Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/17/2005
PI Name:	Pearlman, Howard G. Ph.D.		
Project Title:	Determination of Cool Flame Quenching Distances at Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	howard.g.pearlman@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	216-433-4000
Organization Name:	NASA Glenn Research Center		
PI Address 1:	21000 Brookpark Road		
PI Address 2:	MS 77-1		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:	Howard Pearlman was formerly in the database as a PI at Drexel University. This is the same Howard Pearlman. He is now at GRC working with John McQuillen in PS.		
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	09/16/2002	**End Date:**	09/16/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Brooker, John E	**Contact Phone:**	216-433-6543
Contact Email:	John.E.Brooker@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC3-1003		
Performance Goal No.:			
Performance Goal Text:			
	This ground-based (laboratory and KC-135 aircraft) study aims to (1) isolate the role of conduction heat loss on cool flame quenching and determine cool flame quenching distances (and their associated Peclet numbers) and (2) determine if cool flames can propagate through hot flame quenching distances (and vice versa) for conditions that support both cool and hot flame propagation. Buoyant convection in Earth-based studies affects the transport of heat and species, complicating the determination of the quenching distances. In the absence of buoyant flows, the quenching distances can be more clearly determined. Various hydrocarbon-air premixtures will be tested in the laboratory, and those with chemical induction times that are sufficiently short (e.g., less than 20 s) will be tested aboard NASA's KC-135 microgravity (mg) aircraft. The results (quenching distance, flame speed near the limit of propagation, flame speed as a function of mixture Lewis number, and		

	critical Peclet number) will be tabulated as a function of mixture stoichiometry, temperature, pressure, vessel size, and surface condition. An analytical model that includes thermal and species diffusion will also be developed and compared with the experimental results.
Task Description:	This proposed study fits well within NASA's strategic plan, building a foundation upon which the knowledge gained can then be applied to earth and space-based systems. Specifically, cool flames warrant special attention in reference to aircraft and spacecraft fire safety, since they occur in mixtures with compositions outside of the standard flammability limits and can transition to hot flames under appropriate conditions, i.e., multi-stage ignition. Most recently, the importance of understanding autoignition in static enclosures (e.g., fuel tanks) has been reiterated by the Federal Aviation Administration (FAA) (PR99-18) and the National Transportation Safety Board (NTSB), in response to the Trans World Airlines (TWA) 800 747 center wing tank explosion (FAA and DOT, 1999; NTSB, 1999). While the source of ignition aboard the TWA flight has not been definitively identified, autoignition was considered and not ruled out. It is therefore imperative that we understand the fundamental physicochemical processes and the critical parameters requisite for cool flames and autoignitions to occur to protect against them in undesirable situations. In addition, related research by the PI aims to exploited cool flames to reformulate fuels for fuel cell use as well as reduce industrial costs for gas generation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is relevant to refine chemical kinetic models used to predict the autoigniton of hydrocarbon fuels for use in new engines (homeogeneous charge compression ignition engines), improve our understanding and improve the safety of fuel storage in fuel (aircraft) tanks to suppress low-temperature oxidation reactions, cool flames and autoignition.
Task Progress:	Cool Flames (Numerical Progress) (1) A five-step, skeletal chemical kinetic mechanism including diffusive fluxes of heat and species has been extended to a two-dimensional, axisymmetric flames progataing in narrow channels. This model is able to provide a qualitative understanding of the structure (shape, temperature, and species profile) and stability of cool flames and low-temperature reactions. The role of the Lewis number on flame stability is also explored. In addition, the role of the characteristic vessel size and its role on the laminar to turbulent flow transition was explored (increasing vessel size increases the rise speed of the hot, less dense gas, i.e., Rayleigh number increases). In sufficiently large vessel, the flow is expected to transition to turbulence. This may be the first effort to better understand the role of weak turbulence on cool flame structure, stability and evolution. (2) To provide a quantitative estimate of species concentrations for the proposed space-based Cool Flames Experiment, a PLUG flow reactor numerical computer model is being developed using the Centre national de la recherche scientifique (CNRS) detailed chemical kinetic mechanism for propane and/or n-butane-oxygen premixtures. Once perfected, this model will be used to predict the stable species concentrations appropriate to the proposed test points and used for both safety calculations associated with the exhaust / venting of cool flame products as well as support quantitative gc measurements. The values will be compared with gc stable species measurement accuracies to determine which species can be quantitatively measured in the space-based experiment. Paper Presentation A technical paper entitled “Cool Flames at Earth, Partial, and Reduced-Gravity” was presented at the Eastern States Section of the Combustion Institute at Pennsylvania State University, October 26-29, 2003. It discussed variable-gravity experiments conducted aboard NASA’s KC-135A and the effect of weak convection on cool flame stability, ignition limits and flame shape. Papers In-Progress A technical paper entitled “A Two-Dimensional Gray-Yang model in a rectangular enclosure including natural convection,” is being prepared. Outreach to Undergraduate Students at Drexel University An overview of select experiments and discussion of the research has been incorporated into an undergraduate course (UNIV 101) that Professor Pearlman teaches for incoming engineering students at Drexel University.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Papers from Meeting Proceedings	Fairlie R, Griffiths, J, Hughes K, Pearlman, H. "Cool flames in space: experimental and numerical studies of propane combustion." 30th International Symposium on Combustion: University of Illinois at Chicago IL USA, July 25-30, 2004. 30th International Symposium on Combustion: University of Illinois at Chicago IL USA, July 25-30, 2004. , Jul-2004
Presentation	Pearlman, H., Foster, M. "The Cool Flames Experiment: Recent Results at Reduced and Partial Gravity" 42nd Aerospace Sciences Meeting Jan-2004